

Stoichiometry

Answer key with worked solutions

A. Mole ratios

1. Mole ratios in $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$.

(a) $\text{O}_2 : \text{Al} = 3 \text{ mol} : 4 \text{ mol}$ (b) $\text{Al}_2\text{O}_3 : \text{Al} = 2 \text{ mol} : 4 \text{ mol} = 1 : 2$

2. O_2 for 6.0 mol Al.

$$6.0 \text{ mol Al} \times \frac{3 \text{ mol O}_2}{4 \text{ mol Al}} = 4.5 \text{ mol of O}_2$$

B. Mass calculations

3. MgO from 12.15 g Mg.

$n(\text{Mg}) = 12.15 \text{ g} \div 24.31 \text{ g/mol} = 0.4998 \text{ mol} \rightarrow n(\text{MgO}) = 0.4998 \text{ mol}$ (2 mol MgO : 2 mol Mg). $M(\text{MgO}) = 24.31 \text{ g/mol} + 16.00 \text{ g/mol} = 40.31 \text{ g/mol}$, so mass = $0.4998 \text{ mol} \times 40.31 \text{ g/mol} = 20.15 \text{ g}$

4. O_2 from 24.5 g KClO_3 .

$M(\text{KClO}_3) = 122.55 \text{ g/mol}$; $n = 24.5 \text{ g} \div 122.55 \text{ g/mol} = 0.1999 \text{ mol}$. $n(\text{O}_2) = 0.1999 \text{ mol}$
 $\text{KClO}_3 \times (3 \text{ mol O}_2 / 2 \text{ mol KClO}_3) = 0.2999 \text{ mol}$. Mass = $0.2999 \text{ mol} \times 32.00 \text{ g/mol} = 9.60 \text{ g}$

5. CO_2 from 8.02 g CH_4 .

$n(\text{CH}_4) = 8.02 \text{ g} \div 16.04 \text{ g/mol} = 0.500 \text{ mol} \rightarrow n(\text{CO}_2) = 0.500 \text{ mol}$ (1 mol CO_2 : 1 mol CH_4).
Mass = $0.500 \text{ mol} \times 44.01 \text{ g/mol} = 22.0 \text{ g}$

6. H_2O from 11.0 g C_3H_8 .

$n(\text{C}_3\text{H}_8) = 11.0 \text{ g} \div 44.09 \text{ g/mol} = 0.2495 \text{ mol} \rightarrow n(\text{H}_2\text{O}) = 0.2495 \text{ mol}$
 $\text{C}_3\text{H}_8 \times (4 \text{ mol H}_2\text{O} / 1 \text{ mol C}_3\text{H}_8) = 0.998 \text{ mol}$. Mass = $0.998 \text{ mol} \times 18.02 \text{ g/mol} = 18.0 \text{ g}$

C. Challenge

7. Balance, then find mass of Fe from 100.0 g Fe₂O₃.



(b) $n(\text{Fe}_2\text{O}_3) = 100.0\text{ g} \div 159.70\text{ g/mol} = 0.62617\text{ mol} \rightarrow n(\text{Fe}) = 0.62617\text{ mol Fe}_2\text{O}_3 \times (2\text{ mol Fe} / 1\text{ mol Fe}_2\text{O}_3) = 1.2523\text{ mol}$. Mass = $1.2523\text{ mol} \times 55.85\text{ g/mol} = \mathbf{69.94\text{ g}}$

8. The 2 g : 1 g mistake.

Coefficients compare **moles**, not grams. 2 mol of Mg ($2\text{ mol} \times 24.31\text{ g/mol} = 48.62\text{ g}$) reacts with 1 mol of O₂ (32.00 g); the mass ratio is not 2 : 1.